

Algebraic Identities — Answers

Final answers with a one-line reason. For full methods, see the worked examples in the notes.

Objective

Q1 (b) $a^2 + 2ab + b^2$.

Q2 (b) $a^2 - b^2$. Difference of squares.

Q3 **2ab**. $(a - b)^2 = a^2 - 2ab + b^2$.

Q4 (b) **9801**. $(100 - 1)^2 = 10000 - 200 + 1$.

Q5 **True**. Squaring removes the sign.

Q6 (a) **Both true, R explains A**. $400^2 - 3^2 = 159991$.

Short answer

Q7 (i) $m^2 + 6m + 9$ (ii) $p^2 + 12p + 36$.

Q8 $36x^2 + 60x + 25$. $(6x)^2 + 2(6x)(5) + 5^2$.

Q9 **3025**. $(50 + 5)^2 = 2500 + 500 + 25$.

Long answer

Q10 (i) **2116** (ii) **159991**. $46^2 = (40+6)^2 = 1600 + 480 + 36$; $397 \times 403 = 400^2 - 3^2$.

Q11 (i) $9j^2 + 12jk + 4k^2$ (ii) $4x^2 - 20xy + 25y^2$.

Q12 **16×24 is larger (384 vs 364)**. $14 \times 26 = (20-6)(20+6) = 400 - 36 = 364$; $16 \times 24 = (20-4)(20+4) = 400 - 16 = 384$. The smaller "b" gives the larger product.

Total: 21 marks.